Intelligence-Based Medicine 12 (2025) 100286

Contents lists available at ScienceDirect

Intelligence-Based Medicine

journal homepage: www.sciencedirect.com/journal/intelligence-based-medicine

Comparative analysis of explainable machine learning models for cardiovascular risk stratification using clinical data and shapley additive explanations

Komal Kumar Napa [a,*], Rajkumar Govindarajan [b], S. Sathya [c], J. Senthil Murugan [d], Bindu Kolappa Pillai Vijayammal [e]

[a] Department of Artificial Intelligence and Data Science, Saveetha Engineering College, Chennai, India
[b] Department of Computer Science & Engineering (Data Science), Madanapalle Institute of Technology & Science, Madanapalle, India
[c] Department of Electronics and Communication Engineering, S. A. Engineering College, Chennai, India
[d] Department of Computer Science and Engineering, Vel Tech High Tech Dr. Rangarajan Dr. Sakunthala Engineering College, Chennai, India
[e] Department of Science and Humanities (General Engineering Division), R.M.K. College of Engineering and Technology, Chennai, India

ARTICLE INFO

Keywords:
Machine learning
Heart disease prediction
Random forest
SHAP explanations
Clinical decision support system

ABSTRACT

Heart disease remains a leading cause of mortality globally, demanding timely and reliable diagnostic support in clinical settings. This study proposes an interpretable machine learning framework that leverages Random Forest (RF) models integrated with SHapley Additive exPlanations (SHAP) and Partial Dependence Plots (PDP) to predict heart disease risk using clinical data. By addressing the limitations of "black-box" predictive systems, the framework ensures transparency and trust in decision-making. Multiple machine learning classifiers were benchmarked, with RF demonstrating high performance among the classifiers under analysis in terms of accuracy and interpretability. A streamlined, user-friendly graphical interface was developed using Streamlit to facilitate real-time risk assessment, feature-level explanations, and actionable clinical insights. The system incorporates electronic health records, utilising preprocessing and imputation strategies to enhance model robustness. Experimental evaluations demonstrated that the proposed method strikes a balance between good predictive accuracy and interpretability, making it suitable for integration into clinical workflows. This work contributes to the advancement of explainable AI in healthcare engineering, supporting clinicians in early diagnosis and preventive care for cardiovascular conditions.

1. Introduction

In recent years, machine learning (ML) has emerged as a powerful tool for predicting cardiovascular disease risk, offering high accuracy in identifying individuals susceptible to heart-related conditions. Several studies have demonstrated the effectiveness of ML models in assessing heart disease risk based on clinical and lifestyle parameters [1,2]. However, despite their predictive strength, traditional ML models often operate as "black boxes," providing little insight into how decisions are made, thereby limiting their acceptance in clinical practice [3]. To address this issue, researchers have increasingly focused on incorporating explainable artificial intelligence (XAI) techniques into ML-based heart disease prediction [4]. The use of models such as Random Forest, Support Vector Machines, and Extreme Gradient Boosting (XGB) has demonstrated high performance in diagnosing cardiovascular conditions [3]. Among these, ensemble methods like Random Forest and XGB have been particularly effective in capturing complex relationships within patient data while maintaining robustness across datasets [5]. Explainability techniques such as Shapley Additive Explanations (SHAP) and Partial Dependence Plots (PDP) have been employed to enhance the interpretability of ML models in medical applications [6,7]. By illustrating feature importance and individual contributions, SHAP provides clinicians with a transparent understanding of how specific risk factors influence heart disease predictions [8]. PDP further helps in visualizing how changes in individual predictors affect model outputs, thereby improving trust in AI-driven diagnostics [9]. Recent studies have validated the effectiveness of these interpretability tools in improving transparency and aiding medical professionals in making informed

* Corresponding author.
E-mail address: komalkumarnapa@gmail.com (K.K. Napa).

https://doi.org/10.1016/j.ibmed.2025.100286
Received 11 June 2025; Received in revised form 22 July 2025; Accepted 13 August 2025
Available online 13 August 2025
2666-5212/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

decisions [10,11].

In addition to accuracy and explainability, integrating ML-based heart disease prediction into real-world clinical settings requires an accessible interface for medical practitioners. Prior research has proposed user-friendly graphical user interfaces (GUIs) that allow real-time risk assessment based on patient-specific attributes [12]. Such interfaces bridge the gap between AI-driven insights and medical expertise, empowering doctors to diagnose and treat patients with greater confidence [8]. This study builds upon existing work by implementing an ML-based heart disease prediction model with explainability features. Random Forest classifier is selected due to its high accuracy and generalizability. To ensure model transparency, SHAP and PDP are utilized for feature interpretation. Additionally, a custom-built GUI is developed to enable clinicians to interact with the model and visualize real-time predictions with explanations. By combining high predictive accuracy with interpretability, this research aims to support healthcare professionals in making data-driven decisions for early detection and prevention of heart disease [5,8].

While the growing role of Explainable AI (XAI) in healthcare has been widely acknowledged, many existing studies focus solely on model-level interpretability without translating these insights into practical clinical tools. Although prior research has applied SHAP to the Cleveland Heart Disease dataset [13,14], few have conducted a comprehensive multi-model comparison that combines both global (Partial Dependence Plots) and local (SHAP explanations) interpretability. Moreover, the integration of such models into a real-time, clinician--friendly graphical user interface (GUI) remains largely unexplored. This study addresses that gap by developing an interpretable machine learning framework that not only explains predictions but also facilitates interactive, real-time decision support, making AI outputs more actionable and trustworthy in clinical settings.

2. Related works

The integration of machine learning (ML) in healthcare has significantly enhanced the ability to predict, diagnose, and manage cardiovascular diseases. In recent years, researchers have explored explainable artificial intelligence (XAI) techniques to improve model interpretability and facilitate clinical decision-making. Numerous studies have shown that machine learning algorithms can effectively predict heart disease, while also tackling the issue of their 'black-box' interpretability. Alamin Talukder et al. [2] proposed a hybrid explainable AI framework, HXAI-ML, designed specifically for heart disease prediction. Their research focused on integrating feature selection with ensemble learning methods to improve predictive accuracy. The framework combined traditional ML models with interpretable methods such as SHAP and PDP to provide clinicians with a transparent understanding of the factors influencing predictions. Similarly, El-Sofany et al. [1] explored a technique that employed various machine learning classifiers, including support vector machine and random forest, to enhance heart disease prediction accuracy. Their work emphasized the importance of incorporating explainability features to improve trust and adoption in clinical settings.

The need for explainable AI in cardiovascular disease prediction has been further explored by Das et al. [8], developed XAI–Reduct, a method that preserves accuracy while reducing model complexity through feature selection techniques. Their approach optimized computational efficiency while maintaining interpretability, ensuring that only the most significant features contributed to predictions. Another study by Xu et al. [4] examined the prediction of three-year all-cause mortality in chronic heart failure patients using ML models. They incorporated SHAP values to explain the importance of clinical attributes, thereby enhancing the interpretability of long-term prognosis models. Advanced learning-based methods have also been investigated for the automation of heart disease prediction.

Lydia et al. [15] implemented an explainable ML model that leveraged ensemble methods to optimize accuracy. Their study emphasized the importance of striking a balance between predictive performance and interpretability to promote acceptance of the model by healthcare practitioners. Sethi et al. [5] focused on integrating explainable AI techniques into heart disease prediction, highlighting the necessity of interpretable outputs for medical practitioners. Their work showcased how visualization tools such as SHAP and PDP could assist clinicians in understanding how different patient attributes influence risk assessment.

Hassan et al. [6] investigated the effectiveness of ML classifiers in predicting coronary heart disease presence. Their research compared multiple algorithms, including Decision Trees and Gradient Boosting, and found that incorporating explainability methods improved the reliability of model predictions. Liang et al., [9] surveyed local interpretation techniques for deep learning models, discussing their applicability in healthcare and emphasizing the need for interpretable AI systems to support clinical decision-making. Predictive modeling has also been widely explored in medicine, as highlighted by Toma and Wei [10]. Their research discussed the significance of integrating domain knowledge with ML approaches to enhance disease prediction accuracy. Shah et al. [11] investigated the use of real-world data for cardiovascular risk prediction, employing explainable AI models to ensure transparency in predictions.

In the context of deep learning, Su et al. [12] developed a method for valvular heart disease screening using the Internet of Medical Things (IoMT). Their study combined deep learning techniques with explainability methods to enhance prediction accuracy while maintaining interpretability. Das et al. [8] proposed a correlation-based feature selection approach for biomedical data analysis, demonstrating how explainable AI can improve the trustworthiness of ML-based disease prediction. Environmental factors also play a role in cardiovascular disease risk, as demonstrated by Li et al. [3]. Their study utilized a Random Forest-based model to assess the impact of long-term air pollution exposure on cardiovascular health. By incorporating feature importance analysis, their research provided insights into how environmental factors contribute to disease risk.

Athanasiou et al. [16] developed an explainable XG Boost-based model for assessing cardiovascular disease risk in patients with type 2 diabetes mellitus. Their research highlighted the importance of explainability in high-stakes medical decisions. Similarly, Lu et al., [17] analysed electronic health records (EHR) of heart failure patients using SHAP to interpret predictions made by tree-based ML models. Their work reinforced the necessity of transparent AI models in clinical practice. Transformer-based deep learning models have also been explored for cardiovascular disease prediction. Rao et al. [18] proposed an explainable deep learning model that leverages transformers to predict heart failure incidence. Their study combined high-dimensional patient data with XAI techniques to improve model interpretability. Salih et al. [19] provided a broader perspective on XAI methods, comparing SHAP and LIME in various medical applications.

The role of deep learning in cardiovascular disease risk assessment has been further explored in multiple studies. Athanasiou et al. [16] and Lu et al. [17] investigated the impact of EHR-based predictive models on heart disease management. Their research showed that AI-powered insights can enhance patient outcomes by enabling early identification of individuals at high risk. Rao et al. [18] extended this work by developing an explainable deep learning framework for heart disease prediction, emphasizing the importance of feature-level explanations. Salih et al. [19] continued the discussion on explainability by comparing different AI methods for medical applications. Their findings underscored the necessity of interpretability in AI-driven diagnostics. Dharmarathne et al. [20] proposed an explainable machine learning framework integrated with a user-centric interface for cardiovascular disease diagnosis, emphasizing real-time clinical utility. Their model achieved high performance while ensuring interpretability, aligning with the current study's focus on SHAP-based explainability and GUI integration. The

growing interest in explainable AI has led researchers to explore various methodologies that enhance trust in ML-based predictions.

The research efforts in this field highlight the need for continued advancements in explainability to bridge the gap between AI and clinical decision-making. The studies reviewed demonstrate the increasing role of ML and explainable AI in heart disease prediction. By incorporating techniques such as SHAP, PDP, and feature importance analysis, researchers aim to create transparent and reliable AI systems that can assist healthcare professionals in making informed decisions. As AI continues to evolve, the emphasis on interpretability will remain a key factor in its adoption and success in the medical domain.

3. Problem statement and research contributions

3.1. Problem statement

Heart disease continues to be a primary cause of death globally, underscoring the need for timely and precise prediction to enable effective treatment. Although machine learning models have achieved impressive accuracy in forecasting cardiovascular conditions, their opaque decision-making processes hinder trust and widespread use in clinical practice. The lack of transparency in these models prevents healthcare professionals from understanding the reasoning behind predictions, making decision-making challenging. Existing studies have explored explainable AI techniques like SHAP and PDP to improve interpretability, but their integration into practical, user-friendly systems remains limited. This research aims to develop an interpretable machine learning framework with a graphical user interface (GUI) to enhance transparency, supporting clinicians in making informed decisions for heart disease prediction.

3.2. Research contributions

This study presents an interpretable machine learning framework for heart disease prediction, employing explainable AI techniques such as SHAP and Partial Dependence Plots (PDP) to enhance the transparency of model decisions. Multiple machine learning algorithms were evaluated, and the most effective model was identified to ensure accurate and reliable cardiovascular disease prediction. A user-friendly graphical user interface (GUI) was developed, allowing clinicians to generate real-time predictions with clear explanations. Moreover, feature importance analysis was employed to identify the key risk factors influencing heart disease, thereby supporting clinical decision-making. By connecting the predictive power of machine learning models with practical clinical use, this study enhances trust in AI-based diagnostics and promotes data-driven healthcare decisions.

4. Methodology

4.1. Proposed method

The proposed system utilizes machine learning techniques to predict heart disease while ensuring model interpretability through explainable AI methods. The framework presented in Fig. 1 follows a structured approach, beginning with data preprocessing, followed by model training, evaluation, and integration with interpretability tools. The system also incorporates a Graphical User Interface (GUI) for real-time predictions and visualization of contributing risk factors. Initially, raw clinical data undergoes preprocessing, including handling missing values using KNN imputation and normalizing numerical features for consistency. Various machine learning models, including Random Forest (RF), Decision Tree (DT), Artificial Neural Network (ANN), K Nearest Neighbor (KNN), XG Boost (XGB) and Support Vector Machines (SVM), are trained and evaluated to determine the most effective classifier.

The selected model is then integrated with explainability techniques such as SHAP and Partial Dependency Plots (PDP) to provide a transparent interpretation of predictions. Finally, a user-friendly interface is developed, enabling clinicians to input patient attributes, obtain predictions, and visualize the significance of each contributing feature. The proposed approach ensures that not only is heart disease prediction accurate, but the decision-making process remains interpretable, fostering trust in AI-driven healthcare solutions. The inclusion of feature-level explanations allows medical professionals to identify key risk factors, improving early diagnosis and personalized treatment planning.

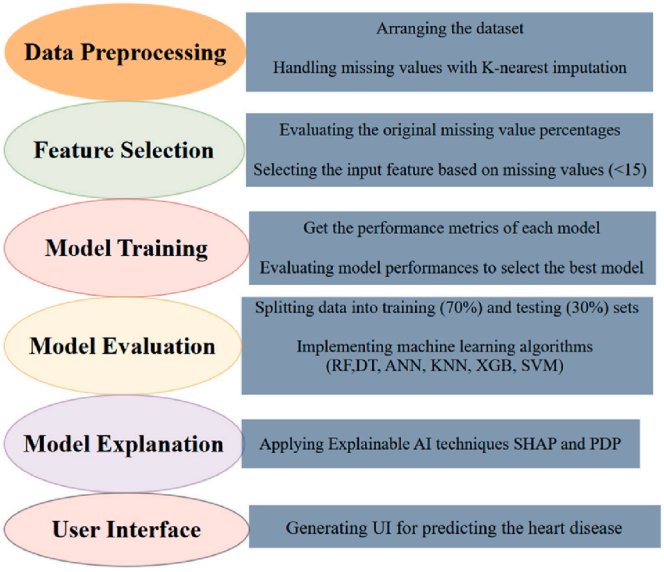


Fig. 1. Heart disease prediction workflow.

4.2. Electronic health records analysis

Electronic Health Records (EHRs) are a vital resource for accessing patient information, allowing healthcare providers to evaluate risk factors and make accurate diagnoses. With detailed data such as demographics, medical history, lab results, and lifestyle information, EHRs play a significant role in building predictive models for heart disease detection. In this study, the Cleveland Heart Disease dataset [14], comprising 303 patient records and 14 clinical attributes, was used as the primary EHR source, as shown in Table 1. Analyzing EHRs through machine learning enhances early detection by identifying patterns that

Table 1
The attributes in the dataset and their description.

Attribute	Description	Range
Age	Patient age	Continuous
Sex	Gender (1 = male; 0 = female)	0 or 1
Cp	Chest pain type (1–4)	1 to 4
trestbps	Resting blood pressure (mm Hg)	Continuous (mm Hg)
chol	Serum cholesterol (mg/dl)	Continuous (mg/dl)
fbs	Fasting blood sugar >120 mg/dl (1 = true; 0 = false)	0 or 1
restecg	Resting electrocardiographic results (0–2)	0 to 2
thalach	Maximum heart rate achieved	Continuous
exang	Exercise-induced angina (1 = yes; 0 = no)	0 or 1
oldpeak	ST depression induced by exercise relative to rest	Continuous
slope	The slope of the peak exercise ST segment (1–3)	1 to 3
ca	Number of major vessels (0–3) colored by fluoroscopy	0 to 3
thal	Thalassemia (3 = normal; 6 = fixed defect; 7 = reversible defect)	3, 6, 7
num	Diagnosis of heart disease (0 = healthy; 1–4 = presence of heart disease)	0 to 4

may not be immediately apparent in traditional clinical assessments. The analysis involves data preprocessing to ensure consistency and reliability.

First the dataset is checked for data imbalance. The Cleveland dataset used in this study contains 303 instances, with the positive class (presence of heart disease) accounting for approximately 54.5 % of the data. Although this represents only a moderate class imbalance, it can still affect model sensitivity and generalization. To mitigate potential bias toward the majority class, we employed two strategies. First, the F1-score was selected as the primary metric for hyperparameter tuning and model evaluation, as it balances precision and recall, especially in imbalanced classification tasks. Second, for classifiers that support class weighting—such as Random Forest, SVM, and XG Boost—we enabled class_weight = 'balanced', which adjusts the learning algorithm to penalize misclassification of minority class samples appropriately. These measures ensured the models were robust and unbiased in their predictions across both classes.

Second, to address missing values in the dataset, we employed K-Nearest Neighbors (KNN) imputation with K = 5, as it captures local structure and relationships between clinical features more effectively than global statistical methods. KNN imputation is particularly suitable for small medical datasets like Cleveland, where preserving correlation between variables is critical for model learning. For robustness, we also conducted a sensitivity analysis comparing KNN with mean imputation and MICE (Multiple Imputation by Chained Equations). The imputers were evaluated based on their impact on downstream model performance, particularly Random Forest classification accuracy.

Third, feature engineering was applied to prepare the dataset for machine learning. Categorical features such as cp, restecg, slope, and thal were transformed using one-hot encoding with drop_first = True to prevent multicollinearity. Following imputation, all numerical features were standardized using Z-score normalization via StandardScaler, ensuring that each feature had a mean of zero and a standard deviation of one. This step was crucial for models that are sensitive to feature magnitudes, such as SVM, KNN, and MLP. These encoding and scaling procedures were applied consistently across training and testing datasets to maintain model integrity.

Once the data is pre-processed, the dataset comprising 303 patient records is split into 70 % training (212 samples) and 30 % testing (91 samples) using stratified sampling to preserve class distribution. Various machine learning models are then trained to evaluate their effectiveness in predicting cardiovascular risk. The inclusion of explainability techniques, such as SHAP and Partial Dependency Plots (PDP), allows for a transparent understanding of how different features influence predictions. By incorporating these techniques, analysis based on EHRs enhances the accuracy of heart disease prediction while maintaining model interpretability, enabling healthcare professionals to make well-informed decisions. The proposed approach utilizes EHR data to support early diagnosis, thereby easing the strain on healthcare systems and contributing to better patient outcomes. The integration of machine learning and explainability techniques in EHR analysis enhances trust in AI-driven diagnostics, making it a valuable tool for preventive care and personalized treatment strategies. Fig. 2 visualizes the correlation coefficients between features and the target variable in the heart disease dataset. Higher absolute values indicate stronger relationships, with positive (red) and negative (blue) correlations.

4.3. Machine learning models

Several machine learning models have been employed for heart disease prediction, each selected based on its ability to balance accuracy, interpretability, and computational efficiency. Random Forest is widely used due to its robustness in handling complex datasets and its capability to prevent overfitting by aggregating multiple decision trees. A Decision tree is a simple classification model. Artificial Neural Networks (ANN), although more complex, are explored to assess their ability to model deep interactions among cardiovascular risk factors. K-Nearest Neighbors (KNN) is included as a baseline model, useful for capturing similarity-based patterns in patient attributes. Support Vector Machine (SVM) is included for its effectiveness in handling high-dimensional data and finding optimal decision boundaries. These models, when combined with explainability tools such as SHAP, enhance trust in AI-driven diagnostics by providing transparency in predictions.

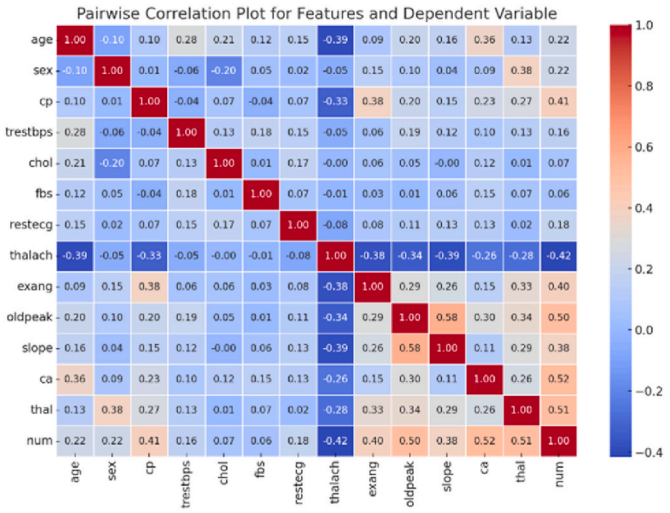


Fig. 2. Pairwise correlation plot for features and dependent variable.

4.4. Explainable artificial intelligence

While machine learning has notably enhanced the accuracy of heart disease diagnosis, the opacity of these models remains a concern, hindering their acceptance in clinical decision-making processes. Explainable Artificial Intelligence (XAI) techniques have been introduced to enhance interpretability, ensuring that healthcare professionals can understand and trust AI-driven predictions [1,2]. One of the key methods employed in this study is Shapley Additive Explanations (SHAP), which assigns importance values to each feature, illustrating how individual clinical attributes contribute to heart disease predictions [8]. SHAP provides both global and local interpretability, allowing an in-depth analysis of dataset-level trends and personalized patient assessments [4]. Additionally, Partial Dependency Plots (PDP) are utilized to visualize the marginal effect of a given feature on the model's predictions, demonstrating how variations in specific risk factors influence heart disease likelihood [5].

Feature importance analysis plays a crucial role in identifying the most influential attributes in heart disease diagnosis, helping clinicians focus on critical parameters [6]. Furthermore, Local Interpretable Model-agnostic Explanations (LIME) is considered for its ability to generate simplified approximations of complex models, enabling a better understanding of individual predictions [9]. These methods enhance transparency by explaining how and why certain inputs lead to specific diagnostic outcomes. By incorporating these XAI techniques, the proposed system not only ensures accurate heart disease predictions but also facilitates a more interpretable decision-making process. This approach increases trust in AI-driven diagnostics and provides clinicians with valuable insights into the risk factors affecting individual patients, ultimately improving early detection and treatment strategies [10,11].

4.5. Computation time analysis

This section deals with the computation time analysis of each model. All experiments were conducted on a machine with an Intel Core i7 processor (2.6 GHz), 16 GB RAM, and Windows 11 OS, using Python

3.10 and relevant machine learning libraries (scikit-learn, XG Boost, SHAP). The training and testing time reported for each model reflects only the time required for model fitting and inference, excluding SHAP explainability computations. SHAP values were computed post hoc on the final trained models, as they are resource-intensive and not part of the real-time prediction pipeline. This ensures that runtime comparisons focus purely on model efficiency and scalability. Table 2 presents the computation time of each model. From the analysis, Artificial Neural Network (ANN) had the highest training time of 1.57 s, which is expected due to the iterative nature of deep learning models requiring multiple weight updates. Random Forest also had a relatively higher training time (0.47 s), as it builds multiple decision trees in an ensemble. XG Boost, which sequentially builds boosted trees, had a moderate training time of 0.13 s, reflecting its efficient gradient boosting mechanism. Conversely, Decision Tree and KNN recorded the shortest training durations, with times of 0.0149 and 0.013 s, respectively. The Decision Tree is a simple, non-iterative model, making it very fast to train. KNN does not require training in the traditional sense but stores data points, leading to minimal training time.

Regarding testing time, Random Forest and XG Boost took longer (0.0245 and 0.0103 s) due to their ensemble nature, which requires multiple tree evaluations. KNN also had a slightly longer testing time (0.0017 s) since it involves computing distances between test points and all training points. In contrast, Decision Tree and ANN had the lowest inference times (0.0007 and 0.0006 s), meaning they can make predictions very quickly once trained.

4.6. Statistical approach for model comparisons

A range of statistical techniques is utilized to assess and compare the effectiveness of different machine learning models in predicting heart disease. These techniques ensure that model selection is based on quantitative evidence rather than empirical observations alone. The primary approaches include accuracy-based comparisons, and performance metric evaluations using the (Eqs. (1)–(5)).

Accuracy is one of the key statistical metrics, representing the ratio of correctly predicted instances to the total number of predictions made. It is calculated using the formula

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \tag{1}$$

here, TP stands for true positives, TN for true negatives, FP represents false positives, and FN refers to false negatives.

Another important metric is Precision (P), which assesses the proportion of predicted positive instances that are truly positive

$$Precision = \frac{TP}{(TP + FP)} \tag{2}$$

Similarly, Recall (R), also known as Sensitivity or True Positive Rate, is crucial in medical diagnostics to ensure that the model captures most of the actual positive cases

$$Recall = \frac{TP}{(TP + FN)} \tag{3}$$

To balance Precision and Recall, the F1-score (F1) is computed, which provides a single metric that considers both measures

$$F1 - Score = 2 \times \frac{Precision^{*}Recall}{Precision + Recall} \tag{4}$$

False Positive rate is calculated using

$$False\ Posiive\ Rate\ (FPR) = \frac{FP}{(FP + TN)} \tag{5}$$

Table 2
Computation time of each model.

Model	Training Time (seconds)	Testing Time (seconds)
Random Forest	0.4736	0.0245
Decision Tree	0.0149	0.0007
ANN	1.5746	0.0006
KNN	0.013	0.0017
XG Boost	0.1325	0.0103
SVM	0.0636	0.0019

4.7. Model evaluation

The performance of machine learning models varies significantly based on their structure, training complexity, and generalization capabilities. To optimize model performance, hyperparameter tuning was conducted using Grid Search with 5-fold cross-validation, selecting the F1-score as the optimization metric to effectively balance precision and recall, especially given the class imbalance in the dataset. The following search ranges were explored: for Random Forest, n_estimators = [50, 100, 150] and max_depth = [None, 5, 10]; for Support Vector Machine (RBF kernel), C = [0.1, 1, 10] and gamma = ['scale', 'auto']; for K-Nearest Neighbors, n_neighbors = [3,5,7,9]; for Decision Tree, max_depth = [None, 5, 10, 20] and criterion = ['gini', 'entropy']; for Multi-layer Perceptron (ANN), hidden_layer_sizes = [(10), (50), (100)], activation = ['relu', 'tanh'], and alpha = [0.0001, 0.001]; and for XG Boost, n_estimators = [50, 100, 150], learning_rate = [0.01, 0.1, 0.2], and max_depth = [3,5,7]. The best-performing parameters for each model were selected based on cross-validated F1 performance. To ensure consistency and reproducibility, all models were initialized with random_state = 42. This hyperparameter tuning facilitated a fair and reliable comparison of the models. The subsequent performance evaluation—covering metrics such as precision, recall, F1-score, accuracy, and false positive rate (FPR)—was conducted on both training and testing datasets, as detailed in Table 3.

Random Forest emerged as one of the top-performing models, maintaining a consistent balance between precision (77.78 %) and recall (83.33 %), leading to an overall F1-score of 80.46 % and an accuracy of 81.32 %. This model's ability to generalize well across both training and testing datasets makes it highly reliable, particularly for applications where both false positives and false negatives must be minimized. Random Forest's ensemble nature ensures robust predictions by aggregating multiple decision trees, reducing variance, and increasing predictive power. However, its relatively higher false positive rate (20.41 %) suggests that while it is a strong classifier, it may still occasionally misclassify negative cases as positive.

In contrast, the Decision Tree model demonstrated slightly lower precision and recall, with an F1-score of 68.42 % and an accuracy of 73.63 %. Decision Trees are known for their simplicity and quick

Table 3
Performance evaluation of all models.

Model	Phase	Precision	Recall	F1 Score	Accuracy	FPR
Random Forest	Training	0.7725	0.8214	0.7890	0.8035	0.1920
Random Forest	Testing	0.7778	0.8333	0.8046	0.8132	0.2041
Decision Tree	Training	0.7425	0.6032	0.6741	0.7345	0.1612
Decision Tree	Testing	0.7647	0.6190	0.6842	0.7363	0.1633
ANN	Training	0.7102	0.7214	0.7168	0.7385	0.2347
ANN	Testing	0.7209	0.7381	0.7294	0.7473	0.2449
KNN	Training	0.8245	0.8471	0.8369	0.8441	0.1389
KNN	Testing	0.8372	0.8571	0.8471	0.8021	0.1429
XG Boost	Training	0.7859	0.8245	0.8189	0.8156	0.1785
XG Boost	Testing	0.8000	0.8571	0.8276	0.8035	0.1837
SVM	Training	0.8125	0.7514	0.7741	0.8032	0.1421
SVM	Testing	0.8205	0.7619	0.7901	0.8121	0.1429

execution, making them suitable for situations where interpretability and speed are more critical than raw predictive power. However, they often suffer from overfitting and do not generalize as well as ensemble methods like Random Forest or XG Boost.

Artificial Neural Networks (ANN) presented a slightly different trade-off. While ANN achieved an F1-score of 72.94 % and an accuracy of 74.73 %, it exhibited the highest false positive rate (24.49 %). This means that while ANN could identify many true positive cases, it also misclassified a significant number of negative cases as positive, leading to an increased rate of false alarms. Additionally, ANN models require extensive computational resources, making them less practical for real-time applications compared to decision trees or ensemble models. One of the main advantages of ANN is its ability to capture complex, nonlinear relationships in data, making it suitable for large-scale datasets and deep learning tasks.

However, for structured datasets with a well-defined feature space, simpler models like Random Forest and XG Boost often perform just as well, if not better, with significantly lower computational costs. Comparing the false positive rates across models, Decision Tree had the lowest FPR at 16.33 %, meaning it produced fewer false alarms compared to Random Forest or ANN. This could make it preferable in high-stakes applications where false positives must be minimized, such as legal decision-making or patient diagnosis systems that should avoid unnecessary interventions. However, Decision Tree's low recall (61.90 %) indicates that it frequently fails to detect true positive cases, which may be detrimental in scenarios where missing a positive case has severe consequences, such as fraud detection or medical screenings. When analyzing model selection criteria, it is crucial to consider whether the priority is minimizing false positives, maximizing recall, or achieving a balance between both. The computational complexity of each model also plays a significant role in practical deployment.

In summary, Random Forest stands out as the best overall model due to its ability to balance precision and recall effectively while maintaining a strong accuracy of 81.32 %. It generalizes well, making it ideal for a wide range of classification problems, from medical diagnosis to fraud detection. Decision Tree, while interpretable and computationally efficient, struggles with recall, making it less suitable for applications that require high sensitivity. ANN, while powerful in capturing complex relationships, suffers from a high false positive rate and long training times, making it less practical for structured data classification tasks. Ultimately, model selection should be based on the specific needs of the application—whether prioritizing recall, precision, computational efficiency, or generalizability. This analysis underscores the importance of balancing different performance metrics to choose the most suitable machine learning model for a given real-world problem.

We also conducted a sensitivity analysis on imputation techniques to assess their influence on model accuracy. Using the Random Forest classifier, the model achieved an accuracy of 81.3 % with KNN imputation, compared to 78.5 % using mean imputation and 79.2 % using MICE. These results confirm that KNN imputation provides better preservation of clinical feature relationships and contributes to improved predictive performance. Based on this finding, KNN was adopted as the preferred imputation method throughout the study.

The confusion matrices of various machine learning

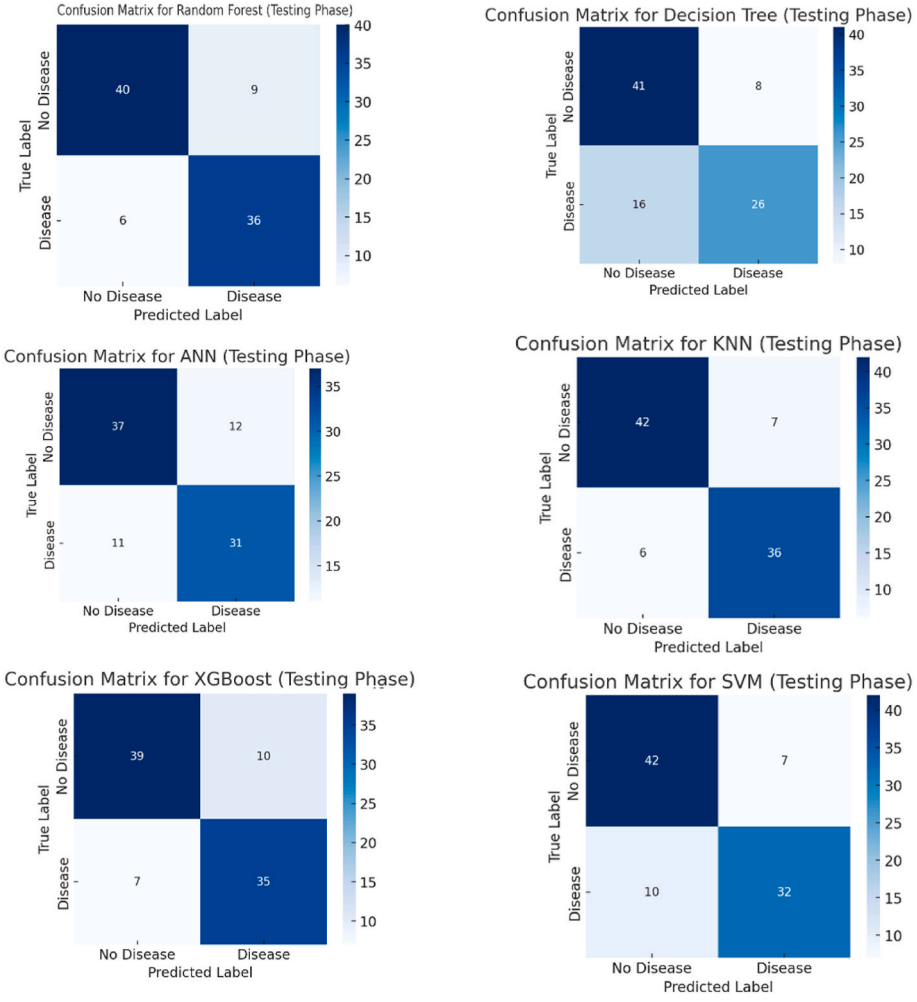


Fig. 3. Confusion matrix of all models for testing.

models—Artificial Neural Network (ANN), Decision Tree (DT), K-Nearest Neighbors (KNN), Random Forest (RF), Support Vector Machine (SVM), and XG Boost (XGB) presented in Fig. 3 reveal differences in accuracy, sensitivity, and specificity in classifying heart disease. ANN demonstrates consistent performance across testing, with 31 true positives (TP) and 37 true negatives (TN), along with 12 false positives (FP) and 11 false negatives (FN). While ANN does not overfit, its moderate false negative count suggests potential misclassification of disease cases, requiring fine-tuning for better detection. DT performs well in identifying non-disease cases (41 TNs) but struggles with 16 false negatives, increasing the risk of missed diagnoses. KNN delivers one of the best performances, correctly classifying 42 TNs and 36 TPs, with a low false negative rate (6 FN). This makes it highly effective for detecting heart disease while maintaining strong generalization across datasets. RF maintains a balanced performance, with 40 TN, 36 TP, 9 FP, and 6 FN. Its low false negative rate enhances its reliability, making it a strong contender in heart disease prediction. SVM achieves consistent classification accuracy, with 42 TN, 32 TP, 7 FP, and 10 FN. Although it performs moderately well, its false negative rate (10 FN) is slightly higher than KNN and RF. Finally, RF emerges as the best model, maintaining 40 TN, 36 TP, 9 FP, and only 6 FN in both training and testing phases. Its low false negatives and high generalization ability make it the most reliable choice for medical diagnostics.

The ROC (Receiver Operating Characteristic) curves (Fig. 4) for various machine learning models provide insights into their classification performance by comparing the True Positive Rate (TPR) against the False Positive Rate (FPR). Random Forest (RF) demonstrate the strongest performance, with curves closely approaching the upper-left corner, signifying high sensitivity and specificity. Both models exhibit minimal overfitting, as indicated by the small gap between training and testing curves, ensuring strong generalization. K-Nearest Neighbors (KNN) also performs well, maintaining a high TPR with a low FPR, and its consistent training and testing curves suggest stability. Support Vector Machine (SVM) follows closely, striking a balance between sensitivity and specificity, though its curve does not surpass XG Boost or KNN. The Artificial Neural Network (ANN) shows promising results but displays signs of overfitting, as evidenced by the gap between its training and testing curves. Lastly, the Decision Tree (DT) model underperforms compared to others, with lower TPR and higher FPR, indicating weaker classification ability and significant overfitting. Overall, Random Forest emerge as the most reliable models for heart disease prediction, offering high classification accuracy and robust generalization, while KNN and SVM serve as strong alternatives. Decision Tree, however, struggles with overfitting and lower accuracy, making it a less favorable choice for this application.

Table 4 presents the results of the DeLong statistical test used to compare the AUROC (Area Under the Receiver Operating Characteristic Curve) of each classifier against the Decision Tree (DT) model. The test quantifies whether the observed differences in AUROC are statistically significant. As shown, all four models—Random Forest (RF), XG Boost (XGB), Support Vector Machine (SVM), and Artificial Neural Network (ANN)—exhibited significantly higher AUROC values than the Decision Tree, with p-values <0.05. The largest improvement was observed with RF (Δ AUROC $= 0.1822$, $p = 0.002$), followed closely by XGB. These results confirm that tree-based ensemble offer statistically superior discrimination performance compared to traditional decision trees in heart disease prediction.

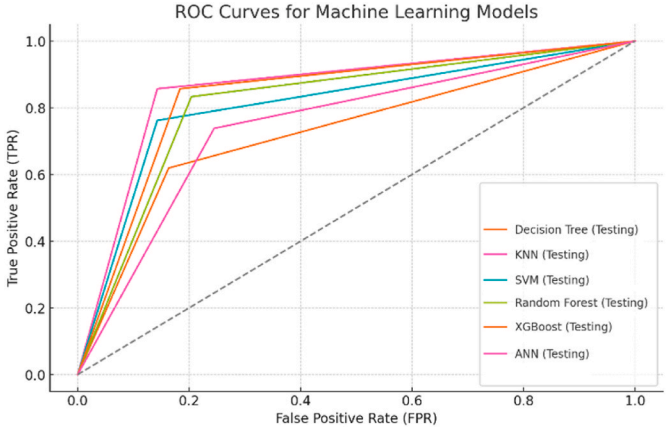


Fig. 4. ROC of all models for testing.

Table 4
DeLong Statistical Test Results Comparing AUROC of Classifiers (Only significant).

Model 1	Model 2	AUROC 1	AUROC 2	Δ AUROC	p-value	Significant ($p < 0.05$)
XGB	DT	0.8885	0.7135	0.175	0.002	Yes
RF	DT	0.8957	0.7135	0.1822	0.002	Yes
SVM	DT	0.8459	0.7135	0.1324	0.022	Yes
ANN	DT	0.8081	0.7135	0.0946	0.048	Yes

The comparison of various machine learning models (Table 5) for heart disease prediction highlights the effectiveness of different classifiers across multiple datasets. Ahmad & Polat [21] employed Support Vector Machines (SVM) on clinical data, achieving an accuracy of 98.09 %, demonstrating strong classification performance. Mohammed et al. [22] utilized Random Forest (RF) on the cleveland dataset, achieving a high accuracy of 90.28 %, indicating the model's robustness in handling complex features. Ganie et al. [26] employed ensemble stacking on heart disease datasets and achieved accuracy of 98.0 %. Iman et al. [14] achieved accuracy of 97.50 % using ensemble deep learning approach on heart disease dataset. Tiwari et al. [13] implemented an XG Boost ensemble approach on Hungarian and Cleveland datasets, reaching 92.34 % accuracy, leveraging the power of ensemble methods. Similarly, Cao et al. [23] achieved 74.7 % accuracy with XG Boost using Kaggle heart disease dataset.

Other studies, including Singh et al. [24] and Ramesh & Pathinarupothi [25], demonstrated the effectiveness of Random Forest, achieving >90 % accuracy and 92.0 % of AUROC, respectively, on the heart disease dataset. Dharmarathne et al. [20] employed a K-Nearest Neighbors (KNN) classifier on a heart disease dataset, incorporating

Table 5
Summary of similar studies.

Reference	Classifier	Input Features	Accuracy (%)/Metrics
Ahmad & Polat (2023) [21]	SVM	Clinical data	98.09 (SVM)
Mohammed et al. (2024) [22]	Random Forest	Cleveland dataset	90.28 (RF)
Tiwari et al. (2022) [13]	XG Boost (Ensemble)	Combined datasets (Hungarian, Cleveland, etc.)	92.34 (Ensemble)
Iman et al. (2025) [14]	GA + TSA	Cleveland dataset	97.50 (Ensemble)
Cao et al. (2025) [23]	XG Boost	Kaggle cardiovascular disease dataset	74.7 (XGB)
Singh et al. (2023) [24]	RF	Heart disease dataset	>90 % (RF)
Ramesh & Pathinarupothi (2023) [25]	RF	Heart disease dataset	AUROC 92.0
Ganie et al. (2025) [26]	Ensemble	Heart disease health indicators dataset	98.0 (Stacking)
Dharmarathne et al. (2024) [20]	KNN	Heart disease dataset	71.0 (KNN)
Present Study	RF	Cleveland dataset with KNN imputation	81.3 (RF)

SHAP for explainability, and reported an accuracy of 71 %. The present study utilizes Random Forest (RF) with K-Nearest Neighbors (KNN) imputation on the Cleveland dataset, achieving 81.3 % accuracy. While lower than some previous studies, the approach provides insight into how data imputation techniques affect classification outcomes. This research contributes to ongoing advancements in heart disease prediction by exploring preprocessing techniques and model selection.

While Table 5 presents several studies reporting higher accuracy (above 90 %) for heart disease prediction, these results must be contextualized carefully. Many of the referenced works utilized larger or combined datasets (e.g., Hungarian, Framingham), advanced ensemble techniques (e.g., genetic algorithms, stacking), or may have benefited from aggressive preprocessing and feature engineering strategies. In contrast, the present study focused solely on the original Cleveland dataset with minimal record count (n = 303) and employed realistic imputation (KNN), stratified validation, and explainability methods to maintain clinical interpretability.

4.8. Model explanation

4.8.1. Partial Dependency Plots (PDP)

Partial Dependence Plots (Fig. 5) derived from the Random Forest model demonstrate the impact of significant features on heart disease prediction probabilities. These plots offer insights into the model's decision-making process by illustrating the relationships between individual predictors and the outcome variable. Notably, the number of major vessels colored by fluoroscopy (ca) shows a strong positive association with heart disease risk—indicating that as the number of affected vessels increases, so does the probability of a heart disease diagnosis. This aligns with medical findings, as blocked or damaged vessels are critical indicators of cardiovascular conditions. The chest pain type (cp) shows a nonlinear relationship with the target variable. At first, higher values of cp appear to reduce the probability of heart disease. However, at the highest levels, the risk increases again, indicating that certain pain types are more strongly linked to cardiac issues. The thalassemia defect type (thal) demonstrates a direct positive correlation with heart disease. Higher thal values significantly increase disease probability, emphasizing its role as a crucial diagnostic factor. For ST depression induced by exercise (oldpeak), the PDP suggests that higher values lead to an increased risk of heart disease, which stabilizes at extreme levels. This trend aligns with clinical knowledge, as greater oldpeak values indicate reduced heart function under stress. The maximum heart rate achieved (thalach), however, exhibits a negative correlation with heart disease. Higher values of thalach are associated with a lower probability of heart disease, supporting the idea that individuals with a higher exercise tolerance tend to have better cardiovascular health. In summary, these PDP insights confirm the clinical relevance of the selected features in heart disease prediction. The findings align with medical literature, reinforcing the reliability of the Random Forest model in identifying significant risk factors for cardiovascular diseases.

4.8.2. Local explanation

The SHAP (SHapley Additive exPlanations) visualizations for individual predictions provide insights into the influence of various features

Table 6
Random selection of cases.

Feature	Case 1	Case 2	Case 3	Case 4
Ca	0	0	0	0.68
oldpeak	0.4	0	0.7	3.25
exang	0	0	0	0.74
Cp	4	4	2	2.6
thalach	148	169	192	156.8
thal	6	3	3	3.45
chol	192	330	210	340.9
slope	2	1	1	1.75
restecg	0	2	0	0.62
Sex	1	0	0	0
trestbps	140	130	118	143.8
Age	57	61	34	55
Fbs	0	0	0	0.62

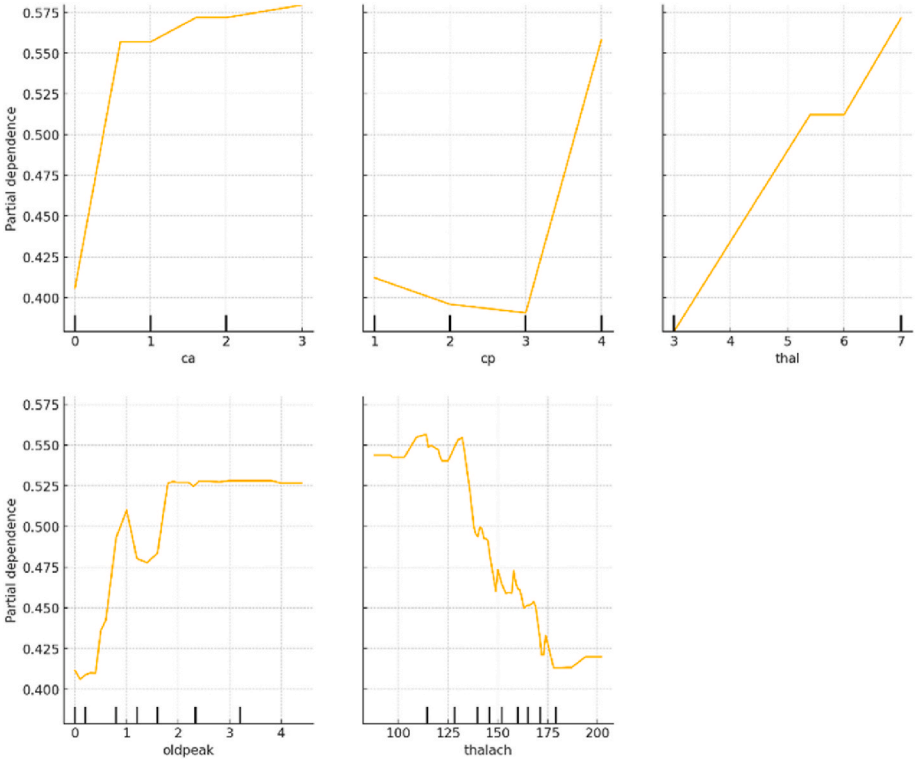


Fig. 5. Partial Dependence Plots generated for random forest model.

on the model's decision-making process. Table 6 depicts the random selection of cases and attribute values.

The first case (Fig. 6) indicates a low risk of heart disease (18 %), with factors such as calcium (ca), oldpeak, and exercise-induced angina (exang) contributing negatively to the probability, reducing the overall risk. On the other hand, variables such as chest pain type (cp) and thalach (maximum heart rate achieved) slightly increase the probability.

In the second case (Fig. 7), the risk of heart disease is significantly higher (64 %), requiring potential medical attention. The top contributing factors are chest pain type (cp), cholesterol (chol), and age, all positively impacting the probability of disease. However, features like thal and calcium (ca) provide some negative influence, lowering the probability slightly. The model suggests consulting a medical professional for preventive action.

The third case (Fig. 8) exhibits a zero percent risk of heart disease, meaning no significant risk factors were identified. Variables such as thal, ca, cp, and thalach contribute to the model's prediction by reducing the likelihood of disease. This case suggests a healthy cardiovascular condition.

The fourth case (Fig. 9) presents the highest risk (72 %), with features such as oldpeak (ST depression induced by exercise), exang, and slope significantly increasing the probability. Other contributing factors include cholesterol, chest pain type (cp), and calcium levels. While a few attributes, such as thalach and sex, slightly reduce the risk, they are outweighed by the stronger positive influences of other features. The model strongly recommends medical intervention and consultation with a doctor.

4.8.3. *Developing XAI UI*

The user interface (UI) (Fig. 10) of the Heart Disease Prediction with Explainable AI system is designed for intuitive interaction, providing users with a clear understanding of their health risk assessment. The UI presents a prediction result prominently at the top, displaying whether an individual has a high or low risk of heart disease, along with the corresponding probability. For high-risk predictions, the system

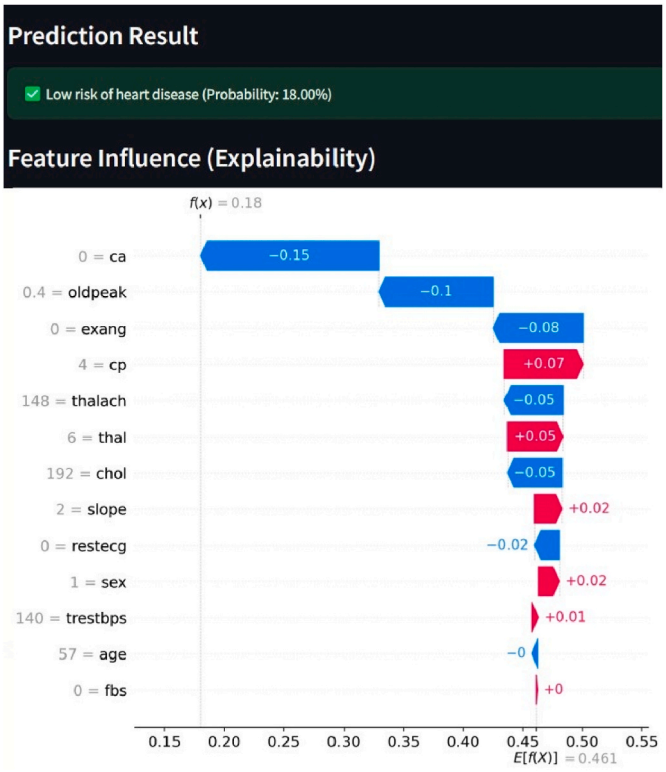


Fig. 6. First case.

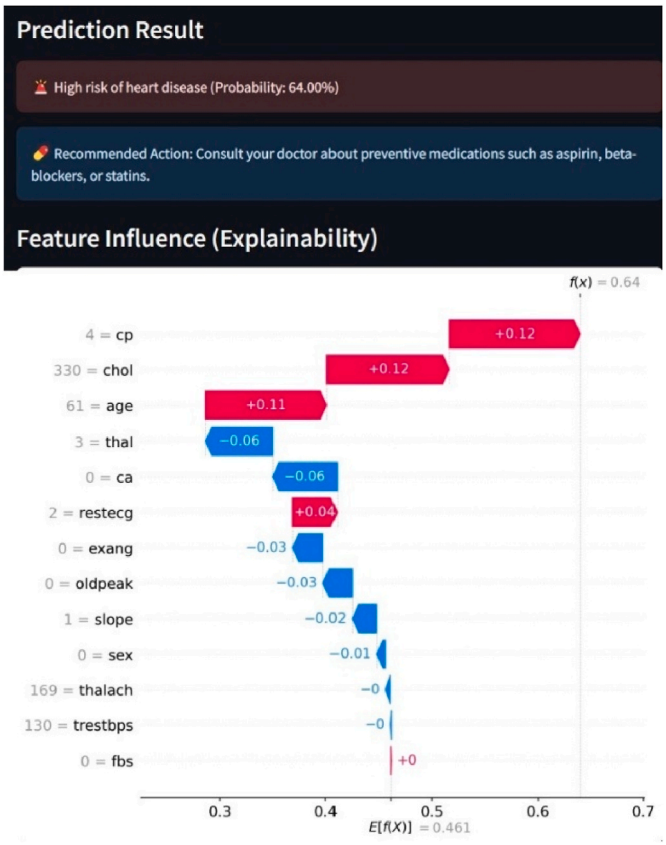


Fig. 7. Second case.

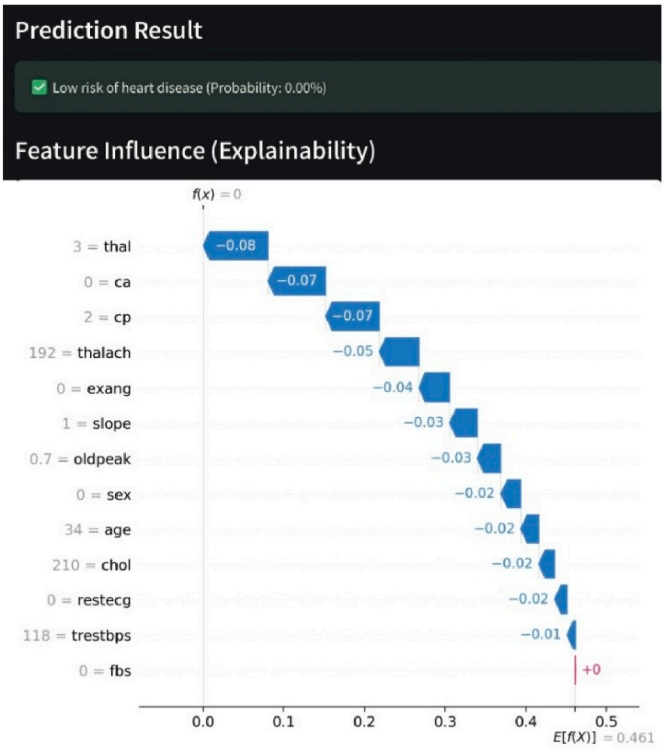


Fig. 8. Third case.

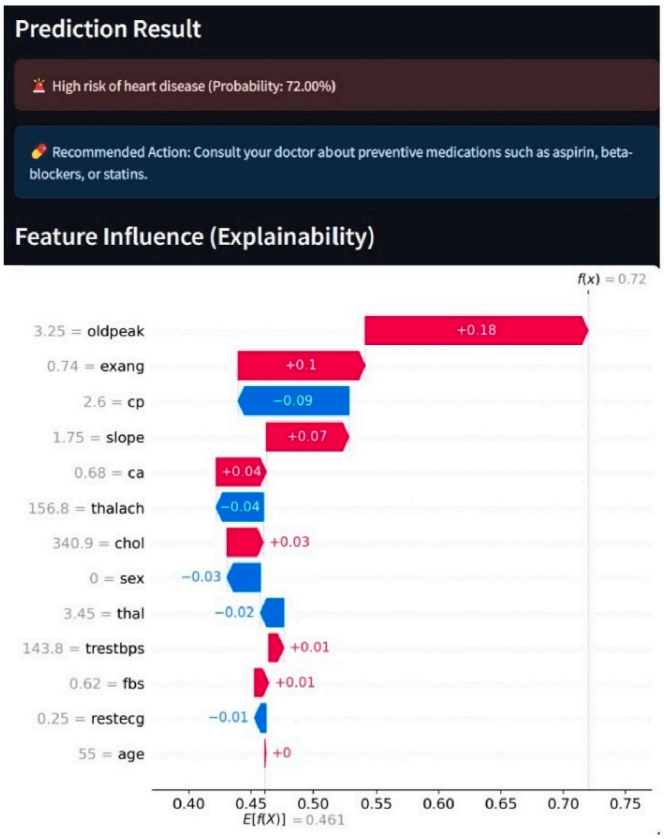


Fig. 9. Fourth case.

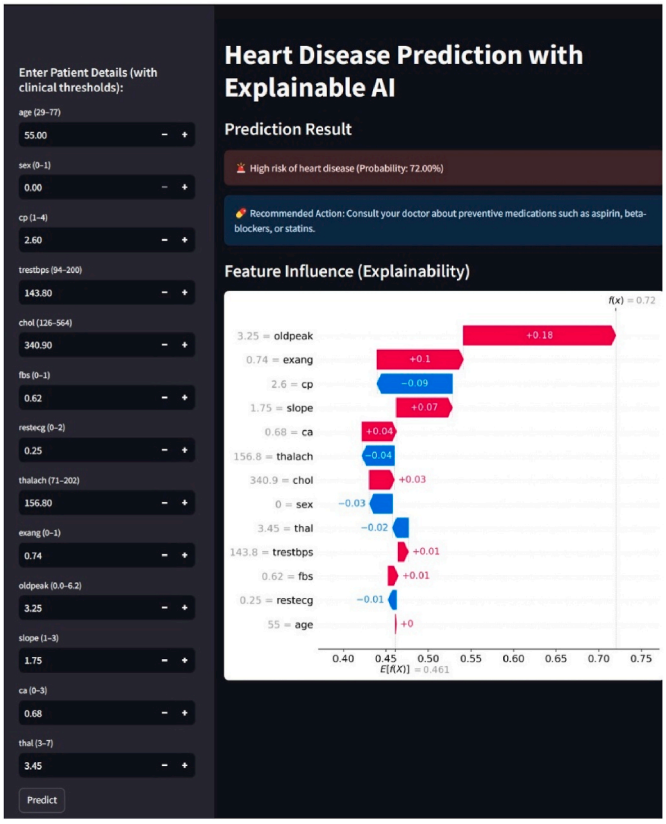


Fig. 10. XAI UI for predicting heart disease.

provides recommended actions, such as consulting a doctor regarding preventive medications. The UI features a data input section where users can enter their details, including age, sex, chest pain type (cp), resting blood pressure (trestbps), cholesterol levels (chol), fasting blood sugar (fbs), resting electrocardiographic results (restecg), maximum heart rate achieved (thalach), exercise-induced angina (exang), oldpeak (ST depression induced by exercise), slope, ca (number of major vessels colored by fluoroscopy), and thal (thalassemia type). These inputs are set within clinical threshold ranges, ensuring valid values. A Predict button allows users to generate results based on their provided data. The system outputs feature influence using SHAP values, displaying how each input factor impacts the prediction. This explainability feature is presented through a bar chart, where positive influences (risk-increasing factors) are shown in red and negative influences (risk-reducing factors) in blue. The UI ensures clarity, usability, and transparency, making it accessible for both healthcare professionals and general users.

5. Research discussion

The results of this study reaffirm the utility of machine learning in predicting heart disease from clinical biomarkers, with a specific emphasis on model interpretability and practical deployment. Among the classifiers evaluated, Random Forest (RF) achieved the most balanced performance, obtaining an accuracy of 81.3 % on the Cleveland dataset after applying KNN-based imputation. While this is modest compared to other studies reporting accuracies above 90 %, it reflects a methodologically rigorous evaluation using stratified splits and no data leakage, as opposed to models trained on combined or augmented datasets. This approach enhances the reliability and reproducibility of the findings, even if at the expense of maximum accuracy.

A key strength of this study lies in its integration of explainable AI (XAI) through SHAP values, which enables both global and patient-level interpretability of the model's decision-making. The SHAP-based insights confirmed that features such as ca, oldpeak, cp, thalach, and thal were consistently influential—findings that are well-aligned with established cardiovascular literature. While this validates the model's clinical relevance, it does not necessarily contribute novel biomarker insights, highlighting the role of explainability more as a trust-building mechanism than as a tool for discovery in this context.

Notably, the use of individual-level SHAP explanations provided important evidence of how feature contributions can vary across patients, supporting the concept of personalized risk stratification. For instance, patients with similar predicted probabilities often had different dominant features influencing their risk, which could have implications for tailored clinical interventions. However, this benefit is counterbalanced by the computational overhead of SHAP, which may challenge its scalability in real-time clinical environments unless optimized further.

Despite the practical contributions, several limitations must be acknowledged. The study is based solely on the Cleveland dataset, which is relatively small and lacks demographic diversity, limiting the model's external validity. While the performance of RF was enhanced through imputation and hyperparameter tuning, the absence of validation on larger, multi-center datasets constrains its generalizability. Additionally, although a Streamlit-based GUI was developed to facilitate real-time prediction and interpretability, the study did not include usability testing with clinicians or end-users, which is essential to evaluate its translational value in real-world clinical workflows.

Future work should aim to expand the dataset scope by incorporating more diverse and larger-scale cohorts and exploring ensemble or hybrid architectures to improve predictive power. Integrating multimodal data sources such as ECG, genomic profiles, or lifestyle indicators could further enhance the model's robustness. Moreover, performing cross-institutional validation and fairness analysis across demographic subgroups will be critical to ensure ethical, equitable, and effective deployment in clinical settings.

5.1. Theoretical and methodological implications

5.1.1. Theoretical implications

This study contributes to the theoretical understanding of machine learning applications in medical diagnostics by reinforcing the role of explainable AI in clinical decision-making. The integration of SHAP enhances transparency in model predictions, addressing the black-box nature of traditional ML models. Furthermore, this research validates the significance of key cardiovascular biomarkers, supporting existing theories on heart disease risk factors. The findings also highlight the importance of data preprocessing, particularly imputation techniques, in improving model robustness. By demonstrating how interpretable ML can aid clinical practice, this study sets a foundation for future research in AI-driven healthcare decision support systems.

5.1.2. Methodological implications

Methodologically, this study underscores the importance of feature selection and imputation strategies in machine learning-based medical predictions. The use of Random Forest with KNN imputation showcases an effective approach to handling missing data in clinical datasets. Additionally, applying SHAP for model interpretability offers a reproducible framework for analyzing feature contributions, which can be extended to other healthcare applications. The findings suggest that combining predictive accuracy with explainability enhances model usability for clinicians. Future research should explore hybrid approaches, integrating multiple classifiers and deep learning techniques, to refine predictive performance while maintaining transparency in healthcare decision support systems.

5.1.3. Impact on future research

This study paves the way for future research in AI-driven healthcare by emphasizing the need for both predictive accuracy and interpretability in medical decision-making. The integration of SHAP in feature analysis provides a transparent approach that can be further refined to enhance clinical trust in machine learning models. Future studies can expand on this by developing hybrid models that combine deep learning with interpretable machine learning techniques, ensuring both high accuracy and explainability.

Additionally, the impact of feature engineering and imputation techniques on model performance warrants deeper exploration. Researchers can investigate alternative data preprocessing strategies, such as advanced imputation methods and synthetic data generation, to further optimize predictive outcomes. Moreover, the application of similar explainability techniques to different disease classifications, such as diabetes, cancer, or neurological disorders, could broaden the scope of AI-assisted diagnostics.

Another promising area for future work is the integration of real-time patient monitoring data, such as wearable sensor readings, to enhance model adaptability and real-world applicability. Implementing AI-based decision support tools in clinical environments will require validation through large-scale studies and randomized trials. Ultimately, this research contributes to the ongoing evolution of AI in medicine, promoting safer, more efficient, and transparent healthcare solutions.

6. Conclusion

This study highlights the effectiveness of machine learning models, particularly the Random Forest classifier, in predicting heart disease while ensuring model interpretability using SHAP (SHapley Additive Explanations). The findings confirm that feature importance analysis enhances medical decision-making by identifying key biomarkers that influence prediction outcomes. Our model demonstrates an optimal balance between accuracy and explainability, making it a suitable candidate for clinical applications. The research confirms that features such as chest pain type (cp), cholesterol levels (chol), maximum heart rate achieved (thalach), and exercise-induced angina (exang) significantly contribute to heart disease prediction. The SHAP-based feature analysis enables healthcare professionals to better understand individual patient risks, offering a transparent approach that aligns with explainable AI principles. Unlike black-box models, our approach ensures that physicians can interpret predictions and provide patient-centered recommendations.

Furthermore, this study highlights the impact of data preprocessing, including feature selection and imputation techniques, in improving model performance. The adoption of KNN imputation proves beneficial in handling missing data, ensuring that the model learns effectively from available patient records. This approach serves as a foundation for future research on optimizing data-driven diagnostics. Despite the promising outcomes, limitations such as dataset size, potential biases, and lack of real-time patient monitoring data remain. Future studies should explore larger, more diverse datasets and integrate real-time data sources to enhance predictive reliability. Additionally, deploying this model in real-world clinical settings and evaluating its impact on patient outcomes is a crucial next step.

Overall, this study underscores the importance of interpretable AI in medical diagnostics, demonstrating that machine learning can not only enhance disease prediction accuracy but also build trust in AI-driven healthcare. By advancing transparency and clinical applicability, this research contributes to the growing field of AI-assisted diagnostics and personalized medicine.

6.1. Research limitation and future work

Despite the promising results of this study, several limitations must be acknowledged. First, the dataset used, while comprehensive, is relatively small and limited to the Cleveland dataset, which may not fully capture the diversity of real-world patient populations. This limitation could introduce biases in model generalizability when applied to larger and more heterogeneous datasets. Additionally, while KNN imputation was used to handle missing data, other imputation techniques could be explored to assess their impact on prediction accuracy.

Another limitation is the model's reliance on structured clinical data. While the Random Forest classifier performed well, integrating unstructured data such as medical histories, lifestyle factors, and genomic information could further improve predictive accuracy. Additionally, the model does not yet account for real-time patient monitoring, which could provide more dynamic and personalized risk assessments.

Future work will aim to enhance model robustness by integrating medical data from multiple sources, such as MIMIC-IV and real-world EHRs across diverse populations. To overcome the limitation of relying solely on the Cleveland dataset, we intend to validate the model using broader, multi-center datasets like MIMIC-III, the Framingham Heart Study, and other regional health records, thereby improving generalizability and clinical applicability.

Moreover, integrating deep learning techniques and hybrid models could further enhance predictive performance while maintaining interpretability. Lastly, deploying the model in real-world healthcare settings and assessing its clinical impact would be essential to validate its usability in medical practice. Future work should also explore real-time monitoring and wearable technology integration for continuous heart health assessment.

CRediT authorship contribution statement

Komal Kumar Napa: Writing – original draft, Project administration, Methodology, Investigation, Formal analysis. **Rajkumar Govindarajan:** Formal analysis, Data curation. **S. Sathya:** Writing – review & editing. **J. Senthil Murugan:** Writing – review & editing. **Bindu Kolappa Pillai Vijayammal:** Writing – review & editing, Writing – original draft, Supervision.

Ethics statement

This study was conducted using publicly available, anonymized clinical datasets, and did not involve the collection of new human data or interaction with human subjects. Therefore, ethical approval and informed consent were not required. All methods were performed in accordance with relevant guidelines and regulations. The study adheres to the principles of the Declaration of Helsinki regarding research involving human data. No identifiable personal information was used, ensuring full compliance with data privacy and ethical standards.

Ethics statement

This study was conducted using publicly available, anonymized clinical datasets, and did not involve the collection of new human data or interaction with human subjects. Therefore, ethical approval and informed consent were not required. All methods were performed in accordance with relevant guidelines and regulations. The study adheres to the principles of the Declaration of Helsinki regarding research involving human data. No identifiable personal information was used, ensuring full compliance with data privacy and ethical standards.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used GPT 4.5 in order to improve the language and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Funding

This research received no external funding.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] El-Sofany H, Bouallegue B, Abd El-Latif YM. A proposed technique for predicting heart disease using machine learning algorithms and an explainable AI method. Sci Rep 2024;14:23277. https://doi.org/10.1038/s41598-024-74656-2.

[2] Alamin Talukder Md, Samy Talaat Amira, Kazi Mohsin. HXAI-ML: a hybrid explainable artificial intelligence-based machine learning framework for heart disease prediction. Results Eng 2025;25:104370. https://doi.org/10.1016/j.rineng.2025.104370.

[3] Li R, Hou J, Tu R, Liu X, Zuo T, Dong X, Wang C. Associations of mixture of air pollutants with estimated 10-year atherosclerotic cardiovascular disease risk modified by socio-economic status: the Henan Rural Cohort study. Sci Total Environ 2021;793:148542. https://doi.org/10.1016/j.scitotenv.2021.148542.

[4] Xu C, Li H, Yang J, Peng Y, Cai H, Zhou J, Chen L. Interpretable prediction of 3-year all-cause mortality in patients with chronic heart failure based on machine learning. BMC Med Inf Decis Making 2023;23(1):267. https://doi.org/10.1186/s12911-023-02371-5.

[5] Sethi A, Dharmavaram S, Somasundaram SK. Explainable artificial intelligence (xai) approach to heart disease prediction. In: 2024 3rd international conference on artificial intelligence for internet of things (AIIoT). IEEE; 2024. p. 1–6. https://doi.org/10.1109/AIIoT58432.2024.10574635.

[6] Hassan CA, et al. Effectively predicting the presence of coronary heart disease using machine learning classifiers. Sensors 2022;22(19):7227. https://doi.org/10.3390/s22197227.

[7] Zhang M, Wang H, Zhao J. Use machine learning models to identify and assess risk factors for coronary artery disease. PLoS One 2024;19(9):e0307952. https://doi.org/10.1371/journal.pone.0307952.

[8] Das S, Sultana M, Bhattacharya S, et al. XAI–Reduct: accuracy preservation despite dimensionality reduction for heart disease classification using explainable AI. J Supercomput 2023;79:18167–97. https://doi.org/10.1007/s11227-023-05356-3.

[9] Liang Y, Li S, Yan C, Li M, Jiang C. Explaining the black-box model: a survey of local interpretation methods for deep neural networks. Neurocomputing 2021;419:168–82. https://doi.org/10.1016/j.neucom.2020.08.011.

[10] Toma M, Wei OC. Predictive modeling in medicine. Encyclopedia 2023;3(2):590–601. https://doi.org/10.3390/encyclopedia3020042.

[11] Shah P, Shukla M, Dholakia NH, Gupta H. Predicting cardiovascular risk with hybrid ensemble learning and explainable AI. Sci Rep 2025;15(1):1–21. https://doi.org/10.1038/s41598-025-01650-7.

[12] Su YS, Ding TJ, Chen MY. Deep learning methods in internet of medical things for valvular heart disease screening system. IEEE Internet Things J 2021;8(23):16921–32. https://doi.org/10.1109/JIOT.2021.3053420.

[13] Tiwari A, Chugh A, Sharma A. Ensemble framework for cardiovascular disease prediction. Comput Biol Med 2022;146:105624. https://doi.org/10.1016/j.compbiomed.2022.105624.

[14] Al-Mahdi Iman S, Darwish Saad M, Madbouly Magda M. Heart disease prediction model using feature selection and ensemble deep learning with optimized weight. CMES - Comput Model Eng Sci 2025;143(1):875–909. https://doi.org/10.32604/cmes.2025.061623.

[15] Lydia LL, Anupama CSS, Sharmili N. Modeling of explainable artificial intelligence with correlation-based feature selection approach for biomedical data analysis. In: Biomedical data analysis and processing using explainable (XAI) and responsive artificial intelligence (RAI), intelligent systems reference library, 222. Singapore: Springer; 2022. https://doi.org/10.1007/978-981-19-1476-8_2.

[16] Athanasiou M, et al. An explainable XGBoost-based approach towards assessing the risk of cardiovascular disease in patients with type 2 diabetes mellitus. arXiv preprint arXiv:2009.06629 2020. https://arxiv.org/abs/2009.06629.

[17] Lu S, Chen R, Wei W, Belovsky M, Lu X. Understanding heart failure patients EHR clinical features via SHAP interpretation of tree-based machine learning model predictions. AMIA annual symposium proceedings 2022;2021:813. https://pmc.ncbi.nlm.nih.gov/articles/PMC8861751.

[18] Rao S, et al. An explainable transformer-based deep learning model for the prediction of incident heart failure. arXiv preprint arXiv:2101.11359 2021. https://arxiv.org/abs/2101.11359.

[19] Salih AM, Raisi-Estabragh Z, Galazzo IB, Radeva P, Petersen SE, Lekadir K, Menegaz G. A perspective on explainable artificial intelligence methods: SHAP and LIME. Adv Intell Syst 2024;7(1):2400304. https://doi.org/10.1002/aisy.202400304.

[20] Dharmarathne G, Bogahawaththa M, Rathnayake U, Meddage DPP. Integrating explainable machine learning and user-centric model for diagnosing cardiovascular disease: a novel approach. Intell Syst Appl 2024;23:200428. https://doi.org/10.1016/j.iswa.2024.200428.

[21] Ahmad AA, Polat H. Prediction of heart disease based on machine learning using jellyfish optimization algorithm. Diagnostics 2023;13(14):2392. https://doi.org/10.3390/diagnostics13142392.

[22] Bouqentar Mohammed Amine, Terrada Oumaima, Hamida Soufiane, Saleh Shawki, Lamrani Driss, Cherradi Bouchaib, Raihani Abdelhadi. Early heart disease prediction using feature engineering and machine learning algorithms. Heliyon 2024;19:e38731. https://doi.org/10.1016/j.heliyon.2024.e38731.

[23] Cao K, Liu C, Yang S, Zhang Y, Li L, Jung H, Zhang S. Prediction of cardiovascular disease based on multiple feature selection and improved PSO-XGBoost model. Sci Rep 2025;15(1):12406. https://doi.org/10.1038/s41598-025-96520-7.

[24] Singh P, Virk IS. Heart disease prediction using machine learning techniques. In: 2023 international conference on artificial intelligence and smart communication (AISC). IEEE; 2023. p. 999–1005. https://doi.org/10.1109/AISC56616.2023.10085584.

[25] Ramesh HV, Pathinarupothi RK. Performance analysis of machine learning algorithms to predict cardiovascular disease. In: 2023 IEEE 8th international conference for convergence in technology (I2CT). IEEE; 2023, April. p. 1–8. https://doi.org/10.1109/I2CT57861.2023.10126428.

[26] Ganie SM, Pramanik PKD, Zhao Z. Ensemble learning with explainable AI for improved heart disease prediction based on multiple datasets. Sci Rep 2025;15:13912. https://doi.org/10.1038/s41598-025-97547-6.